Introduction to multivariate cryptology

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1 Introduction

1.1 Motivation

Polynomial system solving can be applied to cryptography in two ways:

- In security evaluation (cryptanalysis), these algorithms can be used to recover secret information: this is relevant in the analysis of both symmetric and asymmetric schemes.
- Constructively, the NP nature of polynomial system solving enables one to design encryption and signature schemes using trapdoor constructions and zero-knowledge protocols.

The first item is usually referred to as "algebraic cryptanalysis". The second item is referred to as "multivariate cryptography".

Multivariate cryptography is relevant for post-quantum cryptography, as there currently exist no quantum algorithm with a significant advantage over classical algorithms to solve polynomial systems, as opposed to discrete logarithm and factorization.

1.2 National and international standardisation of post-quantum cryptography

The progresses in practical quantum computing suggests that quantum algorithms should be taken into account when studying the security of a cryptographic algorithm. As the main cryptographic algorithms used in modern protocols (such as TLS, SSH, . . .) rely on schemes that are not safe in a quantum model (RSA/DSA/ECDSA/...), post-quantum algorithms must be agreed upon for digital communications. A post-quantum cryptographic algorithm is a classical algorithm ran on a classical computer that is not efficiently breakable by a quantum algorithm.

This vague definition voluntarily hides both the precise definition of efficient quantum algorithm, and also the required size of the quantum computer.

The main standardisation effort has been concentrated around the NIST (National Institute for Standards and Technology, an American agency) since 2017, with a first set of PQC algorithms standardized in 2022 (3 signature schemes and 1 key encapsulation mechanism, which is roughly an encryption scheme). The competition is however not over, as NIST has requested additional signature

schemes avoiding the use of classical lattice-based assumptions and targetting different efficiency tradeoffs.

Other recent standardisation efforts are the Chinese CACR PQC competition and the on-going the Korean KpqC. Many national agencies have published guidelines for the PQC transition: in Europe, see for instance ANSSI [1] and BSI [**?**] .

1.3 Cryptography

We recall some definitions.

Definition 1.1. *A cryptographic algorithm is* asymmetric *(or* public-key*) if the parties communicating do not share the same information. We consider two main examples:*

- *Digital signature schemes enables Bob, using Alice's public key, to verify that a message received was signed using Alice's secret key.*
- *Encryption schemes enables Bob, using Alice's public key, to produce a ciphertext that can only be decrypted using Alice's secret key.*

For $\lambda \in \mathbb{N}$ (usually $128, 192$, or 256), a cryptographic algorithm achieves λ bits of security if compromising any secret information requires less than 2^λ "elementary" operations.

Notice that this definition does not encapsulate quantum algorithms. In practice, standardisation efforts use a reference: the hardness of breaking a well-established algorithm (usually AES 128/192/256), along with a specific set of constraints for quantum computing. We will not discuss quantum algorithms in this course. The "elementary operation" metric is here defined loosely, in practice NIST [**?**] considers classical logical gates.

We define a signature scheme more precisely:

Definition 1.2. *A signature scheme is defined by the following algorithms:*

- KeyGen *takes as input an integer λ (the security level) and returns a pair $(\mathcal{P}, \mathcal{S})$ which is a random*[1] *corresponding public and private key pair of the scheme.*
- Sign *takes as input a private key $\mathcal{S}$, a message $\mathcal{M} \in \{0,1\}^*$ and returns $\mathcal{S}$, a random*[1] *signature of the message.*
- Verify *takes as input a public key $\mathcal{P}$, a message $\mathcal{M} \in \{0,1\}^*$ and a signature $\mathcal{S}$, and returns* True *if the signature is valid and* False *otherwise.*

The signature scheme is deemed secure if with overwhelming probability over the choice of $\mathcal{P}$, producing a couple $\mathcal{M}, \mathcal{S}$ that is accepted by Verify *for $\mathcal{P}$ requires at least 2^λ "elementary operations".*

Note that the definition of security provided in the previous definition is extremely simplified and should only serve as intuition.

[1] in practice, the algorithm is usually deterministic but takes as input a (random) seed

1.4 Notations

Let $q = p^e$ for p prime and $e \in \mathbb{N}_{>0}$. Let $\mathbb{F}_q$ denote the finite field with q elements. p is the characteristic of $\mathbb{F}_q$. Vectors are assumed to be column vectors and are denoted by bold letters: $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{o}, \ldots$. Matrices are denoted by capital letters, and transposition of a matrix is written A^T. For a matrix F_k, the coefficient at position i, j is noted $f_{i,j}^{(k)}$. The kernel of a matrix A is denoted by $\ker(A)$ and is a right kernel: $\boldsymbol{x} \in \ker(A) \iff A\boldsymbol{x} = 0$. Given a field $\mathbb{F}$ and an integer n, we denote by $\mathbb{F}[x_1, \ldots, x_n]$ or $\mathbb{F}[\boldsymbol{x}]$ the polynomial ring of $\mathbb{F}$ in n indeterminates. If I is an ideal of $\mathbb{F}[\boldsymbol{x}]$, the variety defined by this ideal is noted $\mathbb{V}(I) = \{\boldsymbol{x} \in \overline{\mathbb{F}}^n, \forall p \in I, p(\boldsymbol{x}) = 0\}$. The restriction of a function f to a set E is denoted by $f_{|E}$. The canonical basis of the vector space $\mathbb{F}_q^n$ is noted $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$. For a given monomial ordering $\prec$, the leading term of a polynomial p is noted $\mathsf{LT}_{\prec}(p)$. A polynomial is homogeneous of degree d if all its terms have degree d.

1.5 Multivariate Quadratic Cryptography

The number of monomials of degree (exactly) d in n variables is

$$\binom{n+d-1}{d}$$

Therefore, the number of coefficients required to describe a dense polynomial system of m polynomial equations of degree d in n variables is

$$m\binom{n+d-1}{d}$$

This quantity grows quickly when d grows: therefore, it would be very costly to transmit polnyomial systems of large degree for cryptographic purposes. For this reason, we focus on the smallest non-linear systems, those of degree two. In this case, the "size" of a quadratic polynomial system of m equations in n variables over $\mathbb{F}_q$ is:

$$O\left(m\frac{n(n-1)}{2}\log(q)\right)$$

We are concerned with homogeneous quadratic polynomials, polynomials that only have degree two terms. These can be manipulated easily with square matrices:

$$p(\boldsymbol{x}) = \sum_{i=1}^{n}\sum_{j=1}^{n} p_{i,j}x_ix_j = \boldsymbol{x}^T \cdot \begin{pmatrix} p_{1,1} & \ldots & p_{1,n} \\ \vdots & \ddots & \vdots \\ p_{n,1} & \ldots & p_{n,n} \end{pmatrix} \cdot \boldsymbol{x}$$

The function $\boldsymbol{x} \mapsto P(\boldsymbol{x})$ is called a *quadratic form*. Notice that $x_ix_j = x_jx_i$, therefore it is not very useful to distinguish $p_{i,j}$ from $p_{j,i}$. If q is odd, we can consider the matrix $\frac{1}{2}(P+P^T)$ which is symmetric and defines the same quadratic form. If q is even (but it also holds if q is odd), we can assume that P is triangular: replace $p_{i,j}$ by $p_{i,j} + p_{j,i}$ and $p_{j,i}$ by 0.

We will restrict ourselves to digital signature schemes, but there also exist multivariate encryption schemes.

Definition 1.3. *A* multivariate signature scheme *is an asymmetric cryptographic scheme which uses as public key a collection of quadratic polynomials over a finite field* $\mathbb{F}_q$*. The quadratic polynomials of the public key are noted* $\mathcal{P}(\boldsymbol{x}) = (p_1(\boldsymbol{x}), \ldots, p_m(\boldsymbol{x})) \in \mathbb{F}_q[x_1, \ldots, x_n]^m$*. The associated secret key is a set of polynomials related by linear changes of variables:*

$$\mathcal{P}(\boldsymbol{x}) = S \circ \mathcal{F}(\boldsymbol{x}) \circ A$$

The function $\boldsymbol{x} \mapsto \mathcal{P}(\boldsymbol{x})$ *is called a* quadratic map*. We say that* $\boldsymbol{x} \in \mathbb{F}_q^n$ *is a* signature *for the message* $\boldsymbol{t} \in \mathbb{F}_q^m$ *if*

$$\mathcal{P}(\boldsymbol{x}) = \boldsymbol{t} \iff \begin{cases} p_1(\boldsymbol{x}) = t_1 \\ \vdots \\ p_m(\boldsymbol{x}) = t_m \end{cases} \tag{1}$$

In practice, we wish to sign messages with arbitrary length: $\mathcal{M} \in \{0,1\}^*$. We use the hash-and-sign paradigm (which brings other desirable security properties) using a *cryptographic hash function* $\mathcal{H} : \{0,1\}^* \to \mathbb{F}_q^m$:

$$\mathcal{P}(\boldsymbol{x}) = \mathcal{H}(\mathcal{M})$$

For now, we will stick with the simpler formulation provided in the previous definition, and simply assume that $\boldsymbol{t}$ is always chosen uniformly at random.

This construction is interesting because of the following fact: solving polynomial systems of equations over any finite field is an NP-hard task. Therefore, while verifying a solution of any polynomial system is a polynomial-time task, there exist polynomial systems for which finding a solution is hard. The goal of the cryptographer is two-fold:

- Find such systems for which it is hard to find a solution in general.
- Find "trapdoors": a way to find solutions efficiently with secret knowledge.

Keep in mind that a trapdoor introduces a "crack in the shell" of the object considered: it may be easier to recover the trapdoor knowledge than to solve the polynomial system. This gives the following two main cases to consider when studying the security of a cryptographic trapdoor scheme:

- Key recovery: find the secret key or an equivalent trapdoor.
- Message recovery/forgery: compromize (at least) one message/signature.

We clarify these abstract notions with some selected examples in the next sections.

2 Oil and Vinegar

Oil and Vinegar (OV) is a digital signature scheme introduced by Patarin in 1997 [10]. It was broken by Kipnis and Shamir in 1998 [9]. Its successor, Unbalanced

Oil and Vinegar (UOV), introduced by Kipnis, Patarin, Goubin in 1999 [8], has remained secure and is a candidate in the NIST PQC competition, having moved to the second round of the additional signature competition in 2024.

Most of the ideas introduced in OV are relevant for the study of UOV, therefore we will begin our study with this scheme.

2.1 The trapdoor

Let $m \in \mathbb{N}$ be an integer, let q be a power of a prime and let $n = 2m$. m is the number of polynomials in the public key, n is the number of variables and q is the size of the field considered. This means that applied to OV, Equation (1) is a polynomial system of m equations in n variables defined over $\mathbb{F}_q$.

The key idea of OV is that the secret key of Alice will enable her to solve a linear system to sign a message, while an adversary will be confronted to a quadratic polynomial system. To achieve this, Patarin separates the variables in two groups:

- The *oil variables*: $x_1, \ldots, x_m$.
- The *vinegar variables*: $x_{m+1}, \ldots, x_n$.

The polynomials of the secret key are defined as follows, for $1 \leq e \leq m$:

$$f_e(\boldsymbol{x}) = \underbrace{\sum_{i=1}^{m} \sum_{i=m+1}^{n} f_{i,j}^{(e)} x_i x_j}_{\text{oil} \times \text{vinegar}} + \underbrace{\sum_{i=m+1}^{n} \sum_{i=m+1}^{n} f_{i,j}^{(e)} x_i x_j}_{\text{vinegar} \times \text{vinegar}} \tag{2}$$

The takeaway is that the secret polynomials are **linear in the oil variables** since there are no oil $\times$ oil terms. The matrix form may be more visual, with $f_e(\boldsymbol{x}) = \boldsymbol{x}^T \cdot F_e \cdot \boldsymbol{x}^T$ and

$$F_e = \begin{pmatrix} \mathbf{0}_{m \times m} & F_e^{(1)} \\ F_e^{(2)} & F_e^{(3)} \end{pmatrix} \in \mathbb{F}_q^{n \times n} \tag{3}$$

You can keep in mind that at this stage, $n = 2m$. The full secret quadratic map is then:

$$\boldsymbol{x} \mapsto \mathcal{F}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x}))$$

Recall that to sign a message, Alice must solve a system of the form

$$\begin{cases} f_1(\boldsymbol{x}) = t_1 \\ \vdots \\ f_m(\boldsymbol{x}) = t_m \end{cases}$$

Alice proceeds as follows: she chooses random values $v_{m+1}, \ldots v_n$ for the *vinegar* variables

$$x_{m+1} = v_{m+1}, \ldots x_n = v_n$$

and then she tries to find "good values" for the *oil* variables by solving the following system:

$$\begin{cases} f_1(x_1, \dots, x_m, v_{m+1}, \dots v_n) = t_1 \\ \vdots \\ f_m(x_1, \dots, x_m, v_{m+1}, \dots v_n) = t_m \end{cases} \tag{4}$$

A priori, this system is quadratic, but we detail it using Equation (2):

$$\begin{aligned} f_e(x_1, \dots, x_m, v_1, \dots v_{n-m}) =& t_e \\ \iff \underbrace{\sum_{i=1}^{m} \sum_{i=m+1}^{n} f_{i,j}^{(e)} x_i v_j}_{\text{oil} \times \text{vinegar}} + \underbrace{\sum_{i=m+1}^{n} \sum_{i=m+1}^{n} f_{i,j}^{(e)} v_i v_j}_{\text{vinegar} \times \text{vinegar}} =& t_e \\ \iff \sum_{i=1}^{m} \underbrace{\left(\sum_{i=m+1}^{n} f_{i,j}^{(e)} v_j \right)}_{S_i^{(e)}} x_i + \underbrace{\sum_{i=m+1}^{n} \sum_{i=m+1}^{n} f_{i,j}^{(e)} v_i v_j}_{C_e} =& t_e \\ \iff \sum_{i=1}^{m} S_i^{(e)} x_i =& t_e - C_e \end{aligned}$$

where $S_i^{(e)} \in \mathbb{F}_q$ and $C_e \in \mathbb{F}_q$ are constants

This shows that the system in Equation (4) is a linear system of m equations in m variables, which is expected to have a solution with high probability.

In other words, the special shape of the secret key polynomials enable Alice to efficiently solve any system of the form $\mathcal{F}(\boldsymbol{x}) = \boldsymbol{c}$.

2.2 Key generation, signature, and verification

To obtain the public key, Alice needs to hide the structure of the oil/vinegar variables. Therefore, she "mixes" them by applying a random change of variables. Let $A \in GL_n(\mathbb{F}_q)$ be a random invertible matrix.

The public key is the collection of polynomials

$$\mathcal{P}(\boldsymbol{x}) = \mathcal{F}(\boldsymbol{x}) \circ A$$

In matrix form, this is the following collection of square matrices:

$$(A^T \cdot F_1 \cdot A, \dots, A^T \cdot F_m \cdot A)$$

The secret key is the pair $(A, \mathcal{F}(\boldsymbol{x}))$.

Recall that $\mathcal{P}(\boldsymbol{x}) = \mathcal{F}(\boldsymbol{x}) \circ A$. Therefore, Alice signs a message with the following observation

$$\mathcal{P}(\boldsymbol{x}) = \boldsymbol{t} \iff \mathcal{F}(A \cdot \boldsymbol{x}) = \boldsymbol{t}$$

Alice computes $\boldsymbol{y} \in \mathbb{F}_q^n$ such that $\mathcal{F}(\boldsymbol{y}) = \boldsymbol{t}$, and returns $\boldsymbol{x} = A^{-1}\boldsymbol{y}$. Notice that for any message, there exist many valid signatures, which is a useful property of a signature scheme.

When Bob receives a couple (message, signature) of the form $(\boldsymbol{t}, \boldsymbol{x})$, he checks if the following equation holds:

$$\mathcal{P}(\boldsymbol{x}) \stackrel{?}{=} \boldsymbol{t}$$

If it does, he accepts the signature as valid, and rejects otherwise.

We give below a formal definition of the 3 algorithms defining OV, adopting the matricial representation which is easier to implement:

$\mathsf{KeyGen}(\lambda)$

$m, q = \mathsf{Params}(\lambda)$
$n = 2m$
for $e = 1$ **to** m :
  Sample $F_e \leftarrow_{\$} \mathbb{F}_q^{n \times n}$
  for $i = 1$ **to** m :
    for $j = 1$ **to** m :
      $F_e[i, j] = 0$
$\mathcal{F} = (F_1, \ldots, F_m)$
Sample $A \in GL_n(\mathbb{F}_q)$
$\mathcal{P} = (A^T \cdot F_1 \cdot A, \ldots, A^T \cdot F_m \cdot A)$
return $(\mathcal{P}, (A, \mathcal{F}))$

$\mathsf{Sign}((A, \mathcal{F}), \boldsymbol{t})$

Sample $(y_{m+1}, ..., y_n) \leftarrow_{\$} \mathbb{F}_q^{n-m}$
Solve $(\boldsymbol{y}^T F_1 \boldsymbol{y}, \ldots, \boldsymbol{y}^T F_m \boldsymbol{y}) = (t_1, \ldots, t_m)$
return $A^{-1} \cdot \boldsymbol{y}$

$\mathsf{Verify}(\mathcal{P}, \boldsymbol{x}, \boldsymbol{t})$

return $\mathcal{P}(\boldsymbol{x}) - \boldsymbol{t} == \boldsymbol{0}$

Fig. 1: Oil and Vinegar scheme algorithms

2.3 Cryptanalysis: the Kipnis-Shamir attack

The goal of the cryptographer is to provide an algorithm which ensures security: his dream is to prove formally that it would require at least 2^λ operations to break his scheme. Unfortunately, such a proof is in most cases extremely hard, if not impossible, to come up with. A more realistic hope would be to prove that a random instance of the scheme is secure with high probability, which is already a very strong property, but also out of reach in most cases.

Therefore, the trust in a cryptographic algorithm comes from a long history of resilience against attacks. Finding attacks against cryptographic algorithms is called cryptanalysis.

The most naive attack against a multivariate scheme is to try to solve the system $\mathcal{P}(\boldsymbol{x}) = \boldsymbol{t}$ without knowing the secret key. This is a polynomial system of degree two with m equations and $n > m$ variables. Under some regularity

assumption (if the system is not redundant) which are expected to hold on random instanes, finding a solution of such a system with a Gröbner basis algorithm requires an exponential (in m) number of field operations.

More precisely, an upper bound on the number of arithmetic operations required to solve a quadratic system of M equations in N variables that is semi-regular is given by the formula

$$\binom{N+d}{d}^{\omega}$$

where d is the *degree of regularity*, the index of the first non-positive coefficient in the Hilbert series:

$$\frac{(1-t^2)^M}{(1-t)^N}$$

We present in this section the work of Kipnis and Shamir [9], who have shown that it is in fact possible to retrieve the secret key of any Oil and Vinegar instance in polynomial time. In this entire section, we assume that all the matrices in $\mathcal{F}$ are invertible and symmetric[2].

Their first observation was that the secret shape of $\mathcal{F}$ imposed the existence of some interesting linear subspaces.

Lemma 2.1. *Let $(\mathcal{F}, A), \mathcal{P}$ be a UOV key pair for parameters m, q. Define the following linear subspaces of $\mathbb{F}_q^n$:*

- $\mathcal{O}_y = \mathbb{F}_q^m \times \{0\}^m$*, the "y" oil subspace.*
- $\mathcal{O}_x = A^{-1}\mathcal{O}_y = \{A^{-1} \cdot \boldsymbol{y}, \boldsymbol{y} \in \mathcal{O}_y\}$*, the "x" oil subspace.*

Then it holds that

a. For all $\boldsymbol{x} \in \mathcal{O}_y$, $\mathcal{F}(\boldsymbol{x}) = (0, \ldots, 0)$
b. For all $\boldsymbol{x} \in \mathcal{O}_x$, $\mathcal{P}(\boldsymbol{x}) = (0, \ldots, 0)$

Proof. Proof is left as an exercise. Hint: use Equation (2). □

Notice that by definition of $\mathcal{O}_y$, the first m columns of A^{-1} form a basis of $\mathcal{O}_x$.

This lemma is the first hint at the fact that the set of solutions of the system $\mathcal{P}(\boldsymbol{x}) = (0, \ldots, 0)$ is relevant for the cryptanalysis of OV/UOV. This object is called an algebraic variety, but we will not dive deeper in this topic for now. The curious reader can refer to Cox, Little, O'Shea [6] for a gentle introduction to the topic.

Kipnis and Shamir also define the corresponding "vinegar subspaces", and notice an intruiguing relationship with the key:

[2] If q is odd, we have seen that this is always fine, but if q is even, we must use Coppersmith's trick to justify it [9, Remark above Definition 5]

Lemma 2.2. *Let $(\mathcal{F}, A), \mathcal{P}$ be a UOV key pair for parameters m, q. Define the following linear subspace of $\mathbb{F}_q^n$, the "y" vinegar subspace :*

$$V_y = \{0\}^m \times \mathbb{F}_q^m$$

Then it holds that

a. For all $1 \leq e \leq m$, $F_e \cdot \mathcal{O}_y = V_y$.
b. For all $1 \leq e \leq m$, $F_e^{-1} \cdot V_y = \mathcal{O}_y$.
c. For all i, j, $P_i^{-1} P_j \mathcal{O}_x = \mathcal{O}_x$.

Proof. The proof is again left as an exercise. Hint: Rely on the proof of the previous lemma and use the matrix forms. □

By considering the action of the matrices defining the public and private keys, Kipnis and Shamir have shifted from a quadratic polynomial system to the study of linear maps. This is very succesful in this case, as item c. of the previous lemma proves that $\mathcal{O}_x$ is an invariant subspace of $P_i^{-1} P_j$ for all i, j.

Invariant subspaces of endomorphisms are well studied objects in linear algebra, therefore one should expect at this point that a fast algorithm to compute $\mathcal{O}_x$ using only public information (the public key matrices) should exist. We use the following lemmas to set up the computation in practice:

Lemma 2.3. *Let M be a square matrix, let Q be a polynomial. Then* $\ker(Q(M))$ *is an invariant subspace of M.*

Proof. Let $\boldsymbol{x} \in Ker(Q(M))$. $Q(M)\boldsymbol{x} = 0$ by definition. Therefore $M \cdot Q(M)\boldsymbol{x} = 0 = Q(M) \cdot (M\boldsymbol{x})$ since M commutes with any power of itself. Therefore $M\boldsymbol{x} \in Ker(Q(M))$. □

Lemma 2.4. *Let M be a square matrix, let χ_M be its characteristic polynomial. If $\mathcal{O}$ is an invariant subspace of M of dimension d, there exists Q a polynomial of degree d which is a divisor of χ_M such that $\mathcal{O} = \ker Q(M)$.*

Proof. TODO □

The next lemma is a small improvement over the original formulation of Kipnis and Shamir, which yields an easier complexity analysis:

Lemma 2.5. *Let $(\mathcal{F}, A), \mathcal{P}$ be an OV key pair for parameters m, q. Let $1 \leq i < j \leq m$ be two indices and define $M = P_i^{-1} P_j$.*

The following holds:

a. The characteristic polynomial χ_M is a square.
b. With $\chi_M = r^2$, $\mathcal{O}_x = \ker r(M)$.

Proof. a. Recall that for all $1 \leq i \leq m$, F_i is a square symmetric matrix of the following shape:

$$F_i = \begin{pmatrix} 0 & D_i \\ D_i^T & B_i^{(2)} \end{pmatrix} \in \mathbb{F}_q^{2m \times 2m}$$

It holds that

$$M = P_i^{-1} P_j = A^{-1} \cdot F_i^{-1} \cdot (A^{-1})^T A^T F_j A = A^{-1} F_i^{-1} F_j A$$

Thus the matrices M and $F_i^{-1} F_j$ are similar and share their characteristic polynomial. Notice that the shape of F_i imposes that

$$F_i^{-1} = \begin{pmatrix} C_i & (D_i)^{-1} \\ (D_i^{-1})^T & 0 \end{pmatrix}$$

Therefore:

$$F_i^{-1} F_j = \begin{pmatrix} D_i^{-1} D_j^T & * \\ 0 & (D_i^{-1})^T D_j \end{pmatrix}$$

Recall that the characteristic polynomials of $A \cdot B$ and $B \cdot A$ are equal, and the characteristic polynomials of A and A^T are also equal. Notice that $(D_i^{-1} D_j^T)^T = D_j (D_i^{-1})^T$, therefore the two diagonal blocks have the same characteristic polynomial. Therefore, the characteristic polynomial of M is a square.

b. By Lemma 2.2, $\mathcal{O}_x$ is an invariant subspace of M, therefore by Lemma 2.4 there exists Q a divisor of χ_M of degree m such that $\mathcal{O}_x = \ker Q(M)$. If r is irreducible, then $Q = r$ immediatly.

The original proof of Kipnis-Shamir [9, Theorems 9 and 10] stopped here, and only applied the attack when r was irreducible, one we can prove that the attack still succeeds if r factors further which improves and grealy simplifies the complexity analysis. □

2.4 Exercises

Exercise 21 (Proofs)**.** *Prove Lemma 2.1 and Lemma 2.2.*

Exercise 22 (Equivalent keys 1)**.** *Consider an OV key pair $(A, \mathcal{F}), \mathcal{P}$ defined by the relation $\mathcal{P} = \mathcal{F} \circ A$. Prove that adding a transformation T such that $\mathcal{P} = T \circ \mathcal{F} \circ A$ does not add to the security of the scheme, or in other words that A remains a trapdoor of the public key $\mathcal{P}' = T \circ \mathcal{P}$.*

Exercise 23 (Equivalent keys 2)**.** *Consider an OV key pair $(A, \mathcal{F}), \mathcal{P}$ defined by the relation $\mathcal{P} = \mathcal{F} \circ A$.*

1. *Prove that knowing the first m columns of A^{-1} is enough to sign any message.*
2. *Deduce that knowing any basis of $\mathcal{O}_x$ (see Lemma 2.1) is enough to sign any message.*
3. *Is there a specific basis of $\mathcal{O}_x$ that minimizes the number of coefficients to store ? if yes, how many coefficients does it require ?*

3 Unbalanced Oil and Vinegar

The UOV construction [8] is a straight-forward generalisation of OV: the relationship $m = 2n$ is no longer enforced, instead one imposes $n > 2m$. The Kipnis-Shamir attack and its generalisation to UOV are exponential in $n - 2m$, which enables one to build a scheme able to withstand the Kipnis-Shamir attack. One very important remark is that now that $n > 2m$, there are a lot more solutions to the systems $\mathcal{P}(\boldsymbol{x}) = \boldsymbol{t}$ than previously. More precisely, the algebraic variety $V(I)$ defined by this system has dimension at least $n - m$, and in particular in case $\boldsymbol{t} = (0, \ldots, 0)$, we have $\mathcal{O}_x \subsetneq V(I)$ by a simple dimension observation.

3.1 A geometric alternative definition

The cryptanalysis of OV, along with the exercises of the previous section, have shown the following fact: any basis of $\mathcal{O}_x$ can serve as a secret key. To further compress the secret key information, we can focus on the quadratic form formulation of the key pair, following the work of Beullens [3]. Recall that a quadratic form admits a polar form, a symmetric bilinear form defined by the polarization identity:

$$\mathcal{P}^\star(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2}\left(\mathcal{P}(\boldsymbol{x} + \boldsymbol{y}) - \mathcal{P}(\boldsymbol{x}) - \mathcal{P}(\boldsymbol{y})\right)$$

Using this, we can reformulate the geometric trapdoor $\mathcal{O}_x$ (now simply $\mathcal{O}$):

$$\forall \boldsymbol{x}, \boldsymbol{y} \in \mathcal{O}, \mathcal{P}^\star(\boldsymbol{x}, \boldsymbol{y}) = (0, \ldots, 0) \text{ and } \dim \mathcal{O} = m$$

The signing process can then be described using a random "vinegar vector" $\boldsymbol{v} \in \mathbb{F}_q^n$ and computing a good "oil vector" $\boldsymbol{o} \in \mathcal{O} \subset \mathbb{F}_q^n$. Let $\boldsymbol{x} = \boldsymbol{o} + \boldsymbol{v}$ with $\boldsymbol{o} \in \mathcal{O}$:

$$\begin{aligned}
& \mathcal{P}(\boldsymbol{x}) = (t_1, \ldots, t_m) \\
\iff & \mathcal{P}(\boldsymbol{o} + \boldsymbol{v}) = \boldsymbol{t} \\
\iff & \underbrace{2\mathcal{P}^\star(\boldsymbol{o}, \boldsymbol{v})}_{\text{Linear in } \boldsymbol{o}} + \underbrace{\mathcal{P}(\boldsymbol{o})}_{=0} + \underbrace{\mathcal{P}(\boldsymbol{v})}_{\text{Constant in } \boldsymbol{o}} = \boldsymbol{t} \\
\iff & 2\mathcal{P}^\star(\boldsymbol{o}, \boldsymbol{v}) = \boldsymbol{t} - \mathcal{P}(\boldsymbol{v})
\end{aligned}$$

Rather unsurprisingly, after choosing a random value for $\boldsymbol{v} \in \mathbb{F}_q^n$, this is once again a linear system in m variables (the coefficients of $\boldsymbol{o}$ in a basis of $\mathcal{O}$) and m equations.

There are multiple benefits to this approach: this allows one to completely dismiss the "central map" $\mathcal{F}$ that was as large as the public key. Additionally, one can also throw away all but m columns of A^{-1} by definition of $\mathcal{O}$. Even more, one can keep only an echelon form of these m vectors, of the form $(I_m|B) \in \mathbb{F}_q^{m \times n}$, reducing the secret key information to $(n - m) \times m$ coefficients in $\mathbb{F}_q$.

The next remark is a bit technical but not so convoluted: these observations also enable one to reduce the *public key* size. Indeed, the only relevant property of $\mathcal{P}$ is that for all $1 \leq i \leq m$,

$$(I_m|B) \cdot P_i \cdot (I_m|B)^T = 0 \in \mathbb{F}_q^{m \times m}$$

By defining each P_i with blocks of the appropriate size (to fit the block definition of A^{-1}), one can notice that the above equation can be enforced by specifying only one of three[3] blocks of P_i once the other two are chosen. This enables the parties to generate a large part of $\mathcal{P}$ from a short[4] random seed.

3.2 Algorithms

We give below a first definition of the algorithms involved in the UOV signature scheme, but we omit the public key compression method.

Notice also that the signing process may fail and does not specify precisely how the signing randomness ($\boldsymbol{v}$) is generated, which is crucial to ensure the security of the scheme.

$\mathsf{KeyGen}(\lambda)$

$n, m, q = \mathsf{Params}(\lambda)$
for $e = 1$ **to** m :
  Sample $F_e \leftarrow_{\$} \mathbb{F}_q^{n \times n}$
  for $i = 1$ **to** m :
    for $j = 1$ **to** m :
      $F_e[i, j] = 0$
$\mathcal{F} = (F_1, \ldots, F_m)$
Sample $A \in GL_n(\mathbb{F}_q)$
$\mathcal{P} = (A^T \cdot F_1 \cdot A, \ldots, A^T \cdot F_m \cdot A)$
$\mathcal{O} = (\boldsymbol{o}_1, \ldots, \boldsymbol{o}_m) = (A^{-1})[1 \ldots m]$
return $(\mathcal{P}, \mathcal{O})$

$\mathsf{Sign}(\mathcal{P}, \mathcal{O}, \boldsymbol{t})$

Sample $\boldsymbol{v} \leftarrow_{\$} \mathbb{F}_q^n$
Solve $\mathcal{P}\left(\sum_{1 \leq i \leq m} x_i \boldsymbol{o}_i + \boldsymbol{v}\right) = (t_1, \ldots, t_m)$
return $\sum x_i \boldsymbol{o}_i + \boldsymbol{v}$

$\mathsf{Verify}(\mathcal{P}, \boldsymbol{x}, \boldsymbol{t})$

return $\mathcal{P}(\boldsymbol{x}) - \boldsymbol{t} == \boldsymbol{0}$

Fig. 2: Unbalanced Oil and Vinegar scheme algorithms

4 Security analysis of UOV

The security analysis of UOV (see [4, 5] for recent examples) is concerned with two problems:

[3] Recall that a quadratic form is described by either a symmetric or triangular matrix.
[4] This still yields huge key sizes compared to other post-quantum alternatives.

– Forgery attacks: an attacker tries to obtain a signature without the secret key.
– Key recovery attacks: an attacker tries to obtain an equivalent secret key.

Obviously (why ?), the second problem is harder. We give a short overview of both types of attacks.

4.1 Forgery attacks

A forgery attack against a multivariate quadratic signature scheme solves a polynomial system of m equations in n variables over $\mathbb{F}_q$. The best known algorithms to perform this task on finite fields larger than $\mathbb{F}_2$ are based on Gröbner basis computations.

There are two steps to such attacks:

1. Simplify the positive-dimensional system $\mathcal{P}(\boldsymbol{x}) = \boldsymbol{t}$ $(n > m)$ to a zero-dimensional system $S(\boldsymbol{x}) = 0$ (that is usually smaller: $m' \leq m$ equations and m' variables).
2. Compute a solution of the new system.

The first step is usually very fast and involves linear algebra [12]. The second step is trickier, and usually involves a hybrid approach: it is easier to solve an *overdetermined* system (a system with less variables than equations), therefore the attacker enumerates all possible values of a small subset of variables and solves smaller systems:

For instance, when guessing k variables, the attacker solves at most q^k systems of m' equations and $m' - k$ variables:

$$C = q^k MQ(m', m' - k, q)$$

4.2 Key recovery attacks

Reconciliation A key recovery attack targets $\mathcal{O}$. The most natural approach is to write down a very large polynomial system to compute vectors in a basis of $\mathcal{O}$ that vanish $\mathcal{P}$. This is known as the reconciliation attack [7] [3] Let $\boldsymbol{o}_1 = (1, 0, \ldots, 0, x_1, \ldots, x_{n-m}), \boldsymbol{o}_2 = (0, 1, 0, \ldots, 0, x_{n-m+1}, \ldots, x_{2(n-m)})$. The attacker solves the system:

$$\begin{cases} \mathcal{P}(\boldsymbol{o}_1) = 0 & (\text{ m equations}) \\ \mathcal{P}(\boldsymbol{o}_2) = 0 & (\text{ m equations}) \\ \mathcal{P}^*(\boldsymbol{o}_1, \boldsymbol{o}_2) = 0 & (\text{ m equations}) \end{cases}$$

It is hard to estimate the complexity of this attack (as the system is not semi-regular), but it is always more expensive than the direct attack of the previous section.

Key recovery from one vector On the other hand, this attack showcases that once one vector is found, subsequent vectors of $\mathcal{O}$ are significantly easier to find (but still exponential with this attack for proper choices of n, m). One can prove that in fact, once one vector of the secret subspace $\mathcal{O}$ is found, the entire subspace can be recovered in polynomial time (see for example [2], [11]).

Kipnis-Shamir The generalisation of the Kipnis-Shamir attack is due to Kipnis, Patarin, Goubin in [8]. The idea is that eigenvectors of the same endomorphisms $P_i^{-1}P_j$ have an unusually high probability to belong to $\mathcal{O}$, therefore one repeatedely computes eigenvectors of random linear combinations of the P_i:

$$M = \left(\sum_{1 \leq i \leq m-1} y_i P_i^{-1} \right) P_m$$

$$\exists ? \boldsymbol{x}, M\boldsymbol{x} = \lambda \boldsymbol{x} \textbf{ and } \boldsymbol{x} \in \mathcal{O}?$$

One can show that the expected complexity of this attack is

$$O(q^{n-2m} n^{\omega})$$

Modern UOV parameters The tables below have been taken from [5].

	NIST SL	n	m	$\mathbb{F}_q$	|pk| (bytes)	|sk| (bytes)	|cpk| (bytes)	|sig+salt| (bytes)
`ov-Ip`	1	112	44	$\mathbb{F}_{256}$	278 432	237 912	43 576	128
`ov-Is`	1	160	64	$\mathbb{F}_{16}$	412 160	348 720	66 576	96
`ov-III`	3	184	72	$\mathbb{F}_{256}$	1 225 440	1 044 336	189 232	200
`ov-V`	5	244	96	$\mathbb{F}_{256}$	2 869 440	2 436 720	446 992	260

Fig. 3: UOV parameters in [5]

Parameter set (n, m, q)	Collision $\log_2$	Direct k	Direct $\log_2$	KS $\log_2$	Intersection k	Intersection $\log_2$
`ov-Ip` $(112, 44, 256)$	192	2	**145**	218	2	166
`ov-Is` $(160, 64, 16)$	**143**	12	165	154	3	176
`ov-III` $(184, 72, 256)$	304	4	**218**	348	2	250
`ov-V` $(244, 96, 256)$	400	6	**278**	445	2	312

Fig. 4: Complexity of the attacks according to [5]

Notice that the last figure mentions an attack not presented here, the intersection attack. You can learn about it in [3].

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